LLM4RIM: LEVERAGING LARGE LANGUAGE MODEL FOR RADAR-BASED IN-VEHICLE MONITORING

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ABSTRACT

The growing need for non-invasive in-vehicle monitoring in the intelligent cabin underscores the potential of radar sensing. However, existing radar-based solutions often rely on task-specific approaches, lacking a unified framework across multiple downstream tasks. In this work, LLM4RIM, a large language model (LLM) multi-task fine-tuning framework is proposed for radar-based in-vehicle monitoring (RIM) tasks, including occupancy detection, drowsiness detection, gesture recognition, and vital sign monitoring. By utilizing the Mixture of Experts with Low-Rank Adaptation (MoE-LoRA) approach for fine-tuning, the framework leverages the cross-modal learning capabilities of LLMs to enable efficient multi-task learning. Experiments conducted on four task-specific datasets demonstrate an average accuracy of 93.20%, outperforming the state-of-the-art (SOTA) methods by 13.48%. Our code is released at https://github.com/bupt-uwb/LLM4RIM.

***Index Terms*—** Ultra-wideband radar, wireless sensing, large language model, in-vehicle, multi-task learning

1. INTRODUCTION

In-vehicle monitoring technology, as a key component of the intelligent cabin, aims to achieve comprehensive perception and intelligent interaction with occupants. Although vision-based approaches have emerged as relatively mature and easily implemented solutions, they remain limited by their sensitivity to environmental illumination and potential privacy concerns. In contrast, radar sensing has gained increasing attention for its ability to continuously monitor human motion and physiological states under varying lighting conditions while simultaneously avoiding violations of personal privacy [1].

Extensive research has explored radar-based in-vehicle monitoring (RIM) using signal processing and machine learning for tasks such as occupancy detection, gesture recognition, and vital sign monitoring. Early methods [2, 3] rely on traditional machine learning classifiers with hand-crafted features, but suffer from limited generalization and strong dependence on domain knowledge. Deep neural networks have been introduced to better handle environmental interference and multi-path reflections in vehicles [4–7]. For instance, López et al. [6] infer driver stress using radar-derived vital signs as input to a fully connected neural network (FCNN), while Transformer-based models have been applied to ultra-wideband (UWB) radar for gesture recognition [7]. Although these approaches demonstrate strong performance on individual tasks, most are confined to single-task settings, lacking a unified framework for multi-task radar perception.

In recent years, pre-trained large language models (LLMs) have demonstrated remarkable capabilities in reasoning and transfer learning. Multimodal LLMs (MLLMs) fuse text with modalities like images and speech data, further advancing the field by enabling cross-modal embedding alignment and the learning of unified representations [8–10]. However, due to the difficulty of data acquisition, task-specific datasets of signals are often limited [11], insufficient to support the pre-training of MLLMs. Thus, many studies [12–15] primarily employ parameter-efficient fine-tuning (PEFT) approaches to adapt pre-trained LLMs to signal tasks. ECG-LLM [14] incorporates a task-specific encoder-decoder before and after a frozen LLM to achieve restoration of 12-lead electrocardiogram (ECG) signals, while LLM4CP [15] applies low-rank adaptation (LoRA) to fine-tune LLMs for channel prediction. Given the ability to acquire general language representations and powerful reasoning capability from large-scale data, LLMs show strong potential for multi-task learning in wireless signal tasks [16].

In this work, we propose an LLM multi-task fine-tuning framework for RIM tasks, **LLM4RIM**, as shown in Fig. 1. LLM4RIM introduces the mixture-of-experts (MoE) with LoRA (MoE-LoRA) [17] for RIM multi-task fine-tuning of the pre-trained LLMs. By enabling the sharing of expert weights to capture cross-task knowledge and leveraging expert independence and a gating mechanism to preserve task-specific feature differentiation, it achieves both generalization and specialization across tasks. In addition, LLM4RIM employs a task-specific encoder-decoder to ensure effective cross-domain alignment and map UWB radar signals into the LLM embedding space.

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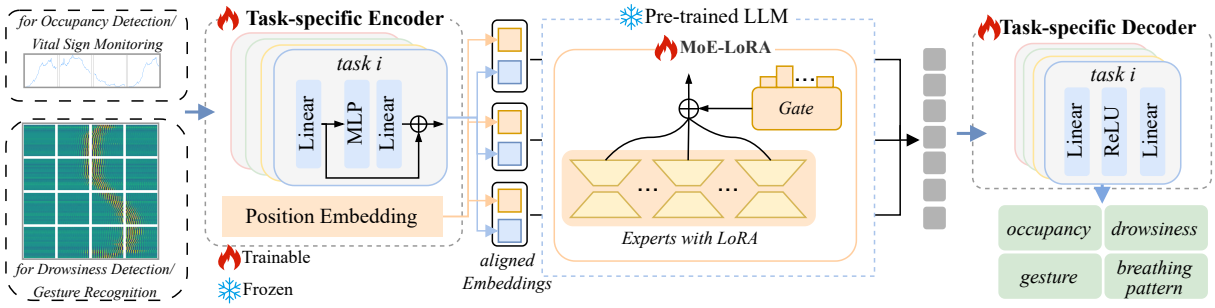


Fig. 1. Architecture of LLM4RIM.

2. METHODOLOGY

2.1. Radar-based In-vehicle Monitoring Tasks

The UWB radar continuously transmits and receives short-duration Gaussian pulses, with the received signal $r(t,\tau)$ sampled into a 2-D matrix R[n,m]:

$$r(t,\tau) = \sum_{i}^{N_{path}} A_{t,i} p(t, \tau - \tau_i) + n(t,\tau) + c(t,\tau) \qquad (1)$$
$$R[n,m] = r(nT_s, nT_f)$$

where, $p(t)$ denotes the transmitted pulse of the UWB radar, $A_{t,i}$ and τ_i represent the scale factor and delay of the received signal, and T_s and T_f correspond to the slow-time and fast-time sampling intervals associated with pulse repetition and time-of-arrival [18], respectively. After preprocessing [19], the received signal predominantly retains features related to human motion and physiological activity.

In this study, we explore the application of LLMs to four representative downstream tasks in RIM: Occupancy Detection **(OD)**, Drowsiness Detection **(DD)**, Gesture Recognition **(GR)**, and Vital Sign Monitoring—specifically, Breathing Pattern Recognition **(BPR)**. These tasks exhibit different emphases on fast-time and slow-time domain features.

OD and BPR tasks rely primarily on 1-D features extracted from either the fast-time or slow-time domain. In OD, the amplitude of UWB signals in the fast time reflects the presence of occupants [20], with the resulting 1-D energy sequence commonly used as an input representation. In terms of BPR, micro-movements of the chest wall modulate the radar phase in the slow time, from which a 1-D respiratory signal can be extracted via downconversion, arctangent demodulation, and phase unwrapping [21]. In contrast, DD and GR tasks rely on 2-D features, requiring continuous monitoring of relatively large-scale motion over time. These tasks inherently involve both fast-time and slow-time information. As a result, neural network models are employed to extract the relevant features directly from the full 2-D matrix, leveraging the joint temporal and spatial structure of the radar signals.

2.2. LLM Multi-task Fine-tuning Framework for RIM

To fully extract the latent features in UWB radar signals across multi-task scenarios and enhance LLM modeling capabilities for signal modalities, a multi-task learning framework named LLM4RIM is proposed. Signals with different shapes are encoded into LLM-compatible embedding vectors. Subsequently, the LLM processes these vectors to generate predictions, and finally, the decoder converts the predictions into task-specific classes.

Task-specific encoder-decoder. Considering radar signals require task-specific balancing between slow-time and fast-time features, LLM4RIM employs parallelized encoder blocks tailored to the input requirements of different tasks. Within each block, the inputs are first partitioned into patches $\{x_1, x_2, ..., x_N\}$, similar to the approach used in Vision Transformer (ViT) [22], and projected into the embedding space of LLMs via a linear layer. Subsequently, the residual multilayer perceptron (MLP) is applied to extract features which are then combined with positional embeddings generated using fixed sinusoidal encoding. The resulting representations are ultimately fed into the LLMs. Regarding the decoder, LLM4RIM employs parallelized task heads to handle the output requirements of different tasks. Each task head consists of MLP followed by a linear layer, which maps the LLM output vectors to the task-specific classes.

Fine-tuning with MoE-LoRA. When extending LLMs to multimodal tasks, full parameter fine-tuning is often prohibitively expensive and prone to overfitting due to limited training data. As a result, LoRA is commonly adopted, which achieves efficient LLM fine-tuning by freezing the original model parameters W_0 and introducing trainable low-rank matrices A and B within selected layers. Thus, for a d-dimensional input vector x, the output is:

$$h = W_0 x + \frac{\alpha}{r} BAx \qquad (2)$$

where r denotes the rank of the low-rank matrices and α is a scale factor, typically set to $2r$.

To further improve representational capacity and parameter efficiency for multimodal applications, [17] combines

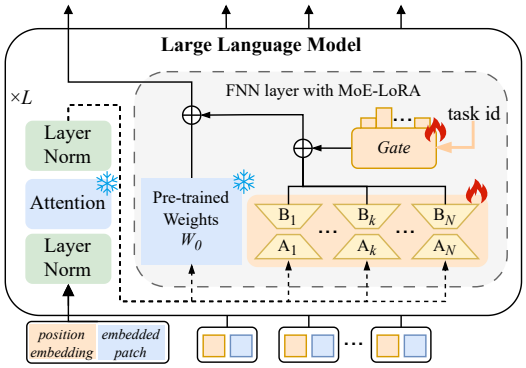


Fig. 2. MoE-LoRA fine-tuning method [17].

LoRA with the MoE principle, as illustrated in Fig. 2. Specifically, experts in the MoE-LoRA layer learn the low-rank matrices and a lightweight gating network assigns task-dependent weights. A Top-K routing strategy selectively activates only a subset of experts, thereby achieving sparse computation and reducing interference across tasks. The forward process for a sample x_t from task t is expressed as:

$$h_t = W_0 x_t + \frac{\alpha}{r} \sum_{k=1}^{N_e} \omega_{t,k} B_k A_k x_t \tag{3}$$

where N_e denotes the number of experts in MoE-LoRA, and $\omega_{t,k}$ represents the weight of an expert in task t. This integration substantially decreases the number of trainable parameters, while enhancing the adaptability of LLMs in multimodal scenarios.

Two-stage Training. The training process is divided into two stages to progressively align radar signals with the semantic space of the pre-trained LLM and enable multi-task joint modeling. In the first stage, the LLM parameters are frozen, and only the encoder and decoder are optimized to learn the mapping between task features and textual representations. In the second stage, the encoder and decoder are frozen, while optimization is applied to the LLM. The LLM is efficiently fine-tuned with MoE-LoRA, reducing computational costs while enhancing cross-task modeling capabilities. To achieve multi-task joint optimization, a weighted composite loss function is employed:

$$\text{loss} = \sum_t \omega_t f_{\text{loss},t} \tag{4}$$

where $f_{loss,t}$ denotes the loss function for task t, and ω_t is the corresponding weight used to balance the contribution of each task to the total loss. To ensure effective optimization across tasks during multi-task training, the dynamic weight average (DWA) algorithm [23] is employed, which adaptively adjusts task weights based on the loss trends observed over consecutive epochs.

3. EXPERIMENTS

3.1. Implementation Details

The information of the datasets used in the experiments is shown in Table 1. Publicly available datasets [24, 25] were employed for GR and BPR tasks, respectively, while two dedicated datasets were collected via X4M03 UWB radar for OD and DD tasks due to the lack of public benchmarks. The X4M03 radar operates at a center frequency of 7.29 GHz with a bandwidth of 1.4 GHz, featuring a high sampling rate of 23.328 GS/s and a slow-time sampling of 100 Hz. The effective detection range was configured from 0.2 to 3 m. The OD dataset involved 0–5 occupants under various seating configurations, with the radar mounted on the vehicle's A-pillar. For the DD dataset, data were acquired using a driving simulator simulating three driver states (normal, excited, and drowsy). All datasets were split 8:2 for training and testing.

Table 1. Details of task-specific datasets.

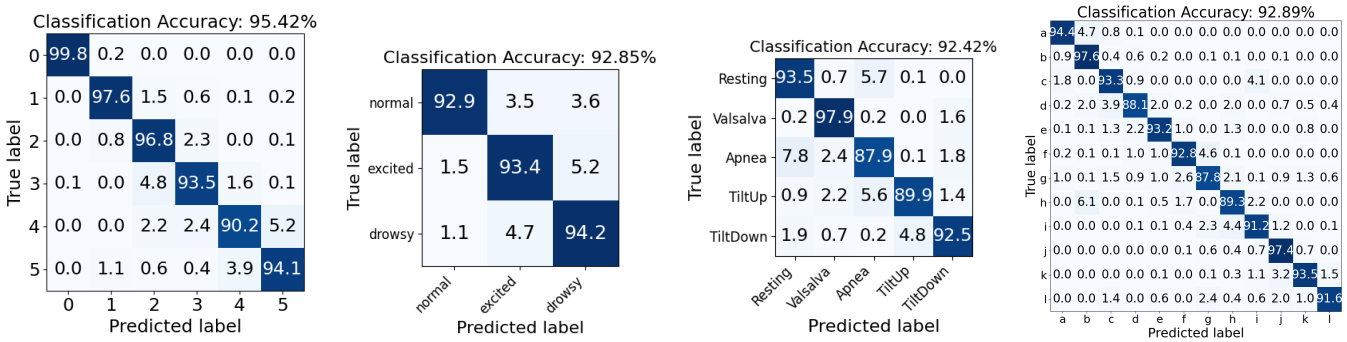
Task	Subjects	Classes	#Samples
OD	5	6	2900
DD	10	3	2550
BPR [24]	30	5	8040
GR [25]	8	12	9500

Model training was conducted using NVIDIA A800 80GB GPUs with CUDA 11.8 and fixed random seeds to ensure reproducibility. The DeepSeek R1-7B [26] and Llama 3-8B [27] models served as pre-trained LLMs for fine-tuning. The architecture employed a LoRA rank of 8 with 8 experts. Training was performed using the AdamW optimizer with an initial learning rate of $1e^{-4}$, over 50 epochs with a batch size of 64. FP16 mixed precision was employed, along with linear learning rate decay and a 10% warm-up phase to enhance stability and efficiency.

3.2. Experimental Results

For comparison, we select some state-of-the-art (SOTA) models [4–7] tailored to each of the four RIM tasks, as detailed in Table 2. Notably, all baseline models are trained independently for each task, in alignment with their respective design objectives and optimization objectives specific to their individual architectures.

The classification result of LLM4RIM in all four RIM tasks is visualized in Fig. 3. The results in Table 2 and Fig. 3 demonstrate that LLM4RIM consistently delivers high performance on all tasks, with the OD task achieving significantly higher accuracy compared to existing methods. Leveraging a shared pre-trained LLM as its backbone and incorporating task-specific knowledge through MoE-LoRA, the proposed framework effectively balances shared representation learning and task-specific adaptability. This further validates the



(a) Occupancy Detection (b) Drowsiness Detection (c) Breathing Pattern Recognition (d) Gesture Recognition

Fig. 3. Classification performance of four RIM tasks with confusion matrix.

Table 2. Performance (%) of different methods in four RIM tasks.

Method	OD	DD	BPR	GR	Avg.
FCNN [6]	67.31	92.10	91.21	62.45	78.27
3DCNN+LSTM [4]	92.07	77.57	77.45	80.11	81.80
CNN+LSTM [5]	82.85	75.30	**93.27**	77.10	82.13
Transformer [7]	74.12	78.00	73.51	**94.50**	80.03
LLM4RIM (DeepSeek R1-7B)	**95.42**	**92.85**	92.42	92.13	**93.20**
LLM4RIM (Llama 3-8B)	**95.03**	**92.71**	90.88	92.89	**92.88**

application potential of LLMs in the field of radar sensing.

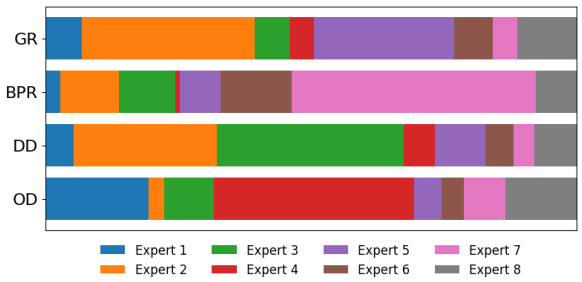


Fig. 4. Expert weights for RIM tasks.

Fig. 4 presents a visualization of expert weight distributions in four tasks. For each task, the lengths of the color-coded bars represent the normalized weights assigned to the corresponding experts, with the sum of weights normalized to 1. It is evident that the contributions of individual experts vary significantly, indicating that the gating network indeed learns task-specific combinations of experts. Notably, the DD and GR tasks exhibit highly similar distributions of expert weights, likely reflecting their underlying similarities. This observation further supports the effectiveness of MoE-LoRA in leveraging shared knowledge to enhance performance on related tasks.

To further investigate the contribution of the LLM module, we conduct a series of ablation studies in which the backbone LLM module is either frozen or entirely removed during training. The corresponding performance outcomes are summarized in Table 3. Across all ablation configurations, we observe a notable degradation in task performance compared to the complete LLM4RIM setup. This consistent performance drop highlights the critical role of the LLM in capturing high-level semantic representations and facilitating effective feature integration across tasks.

Table 3. Ablation study on the backbone LLM module

Method	Average Accuracy
LLM4RIM	**93.39%**
w/o LLM	84.48%
frozen LLM	87.65%
w/o MoE	90.17%

4. CONCLUSION

In this paper, we introduce a multi-task fine-tuning framework specifically tailored for LLMs in RIM tasks. By designing task-specific encoders and decoders, LLM4RIM effectively aligns the radar signal feature space with the semantic space of LLMs, and further utilizes MoE-LoRA for efficient multi-task fine-tuning. This framework offers a novel perspective on integrating multimodal perception with radar signal processing.

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